CS 663: Fundamentals of Digital Image Processing
Assignment 2

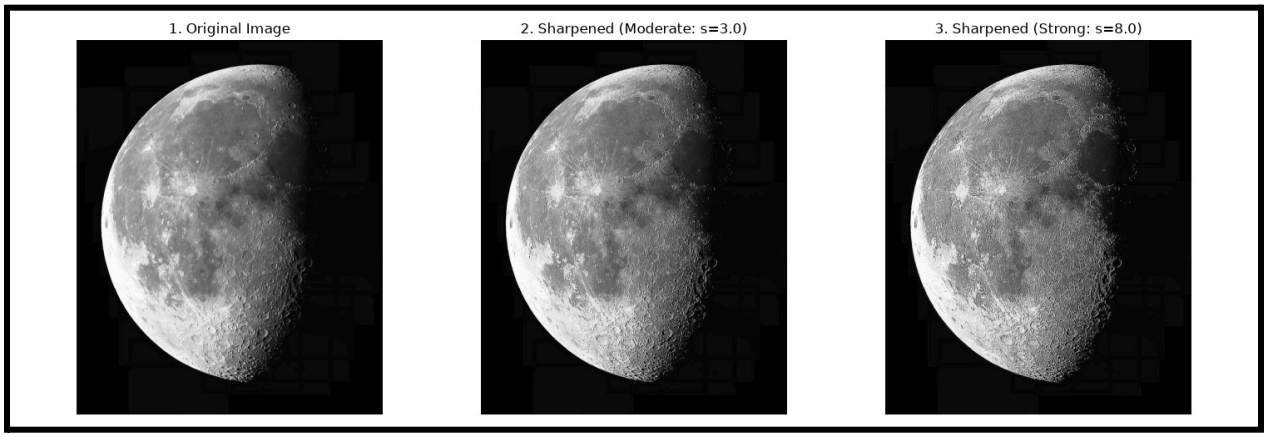
(Filtering)

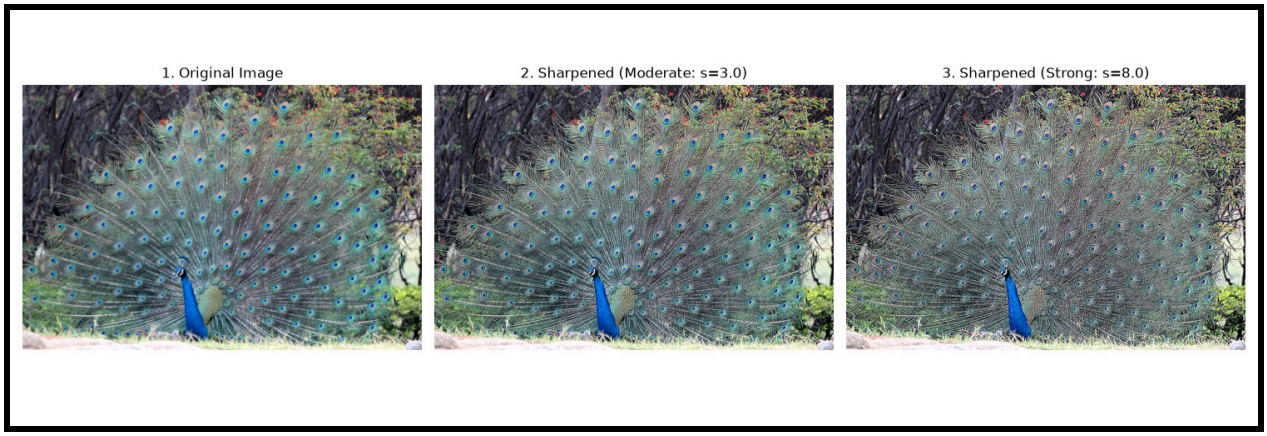
Date: 11th September 2026

By

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1. Image Sharpening



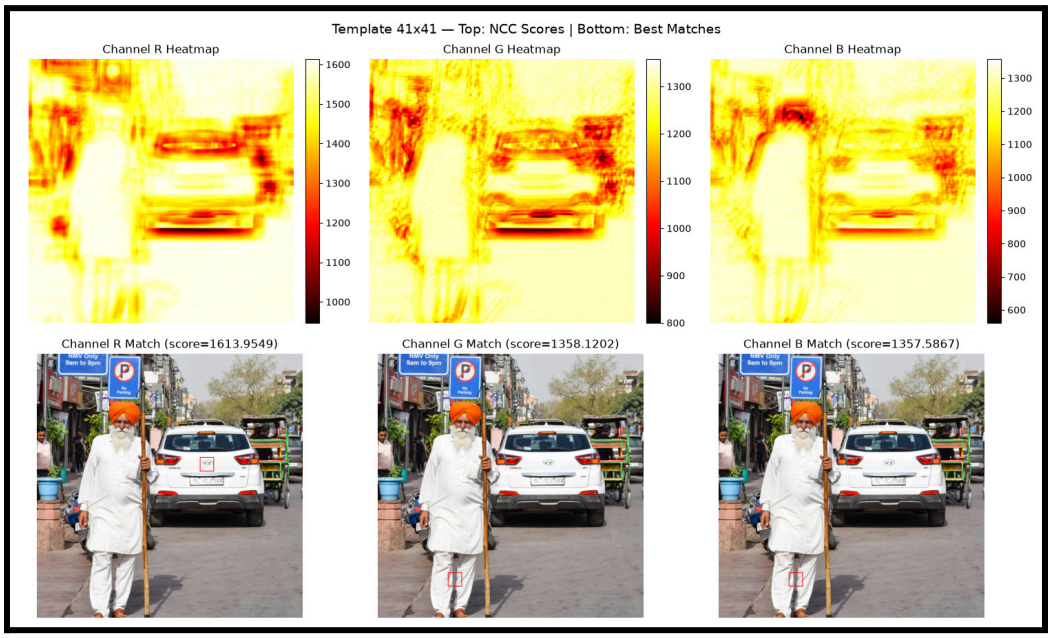


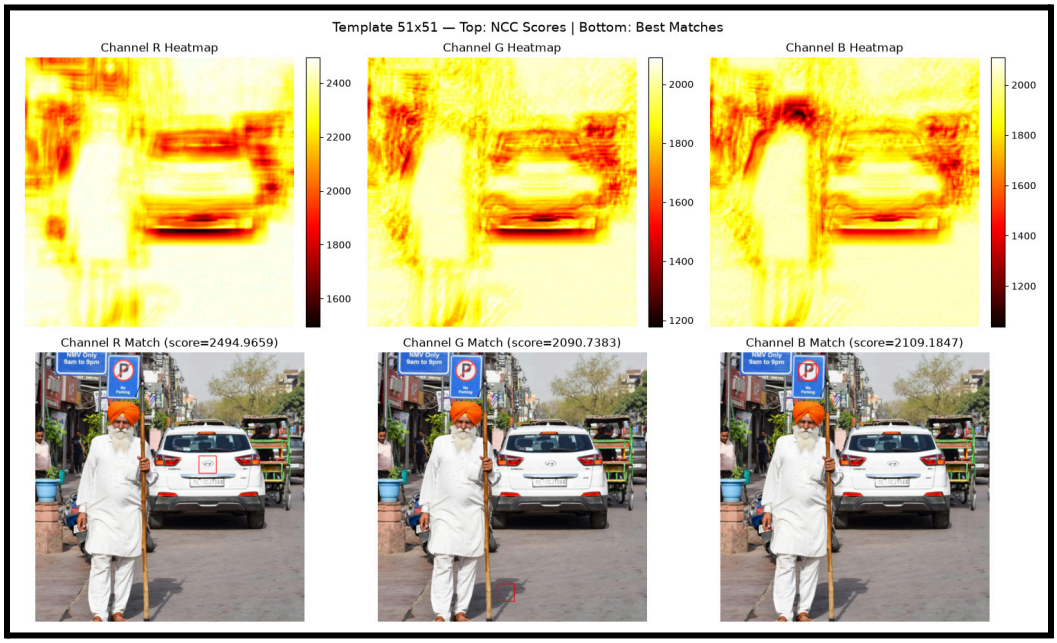


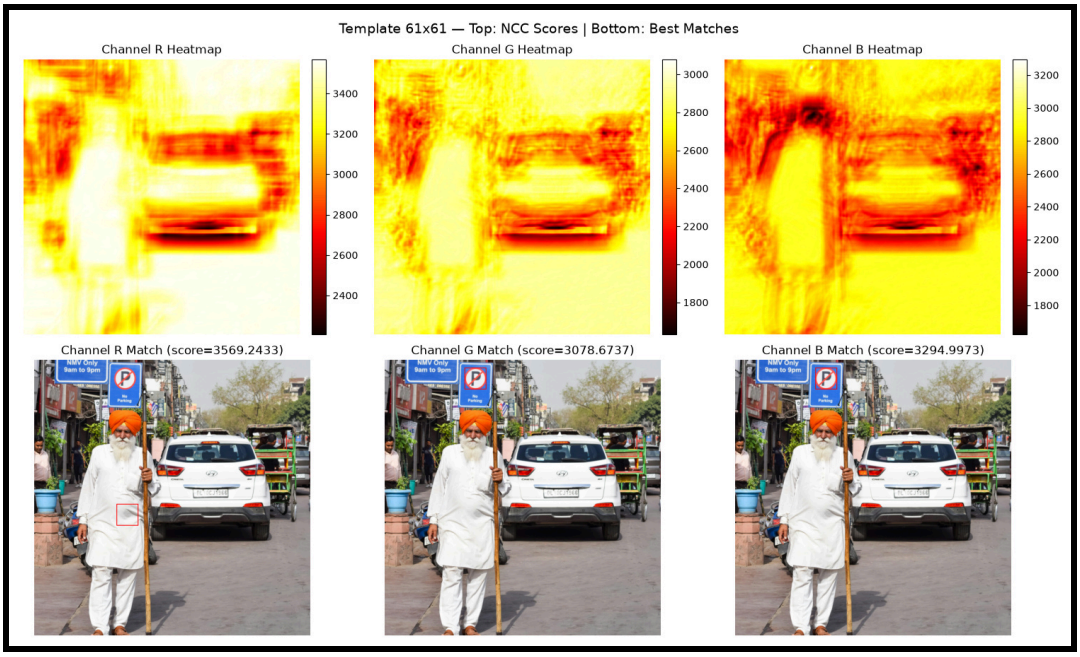
Explanation: When we apply unsharp masking to the images:
1. moon.png: It amplifies the local contrast along the rock formations when s = 3 and the rock features which make the craters and ridges are significantly crisper and more distinct. When s is further increased to 8, even the fine rock features get amplified resulting in a grainy image.

2. peacock.png: It aggressively targets the high-frequency details when s=3 which includes the fine details in the feathers of peacock and making it more prominent. At s=8, the texture becomes extremely dense and the image becomes less blurry as compared to the original image.
3. tiger.png: The sharpening algorithm targets the features like the tiger's fur, its whiskers, and the sharp outlines of the floating leaves. It gives a sharper look to those features when s=3 but when s is increased to 8, the setting gives the wet fur a much harsher but a highly defined look.

2. Template Matching (Normal NCC)







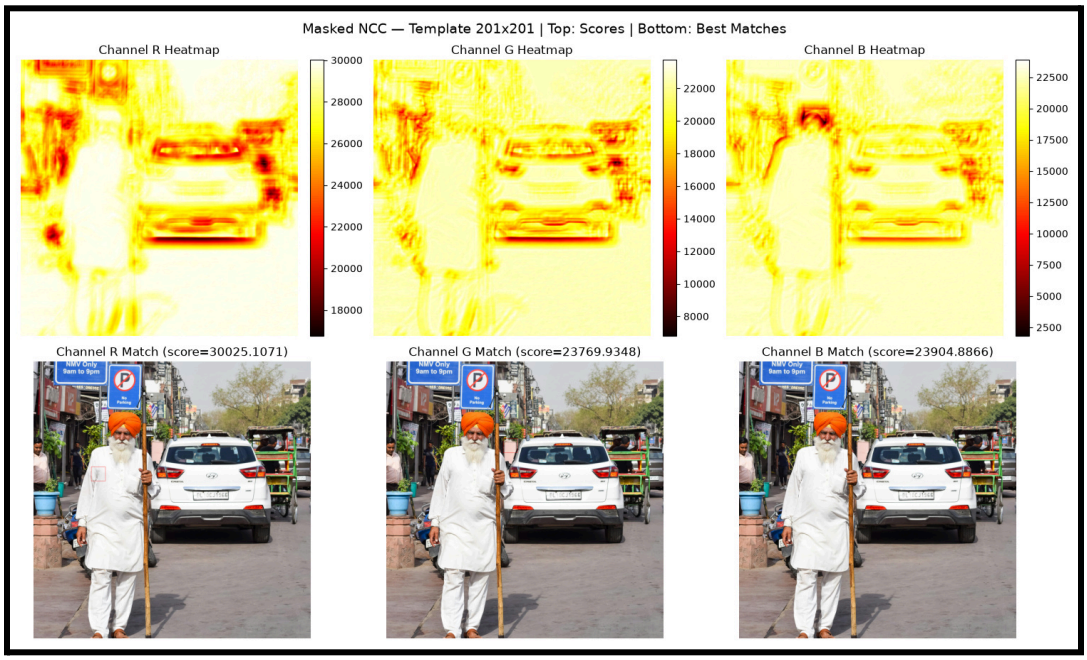
Explanation: When we apply template matching to the image:

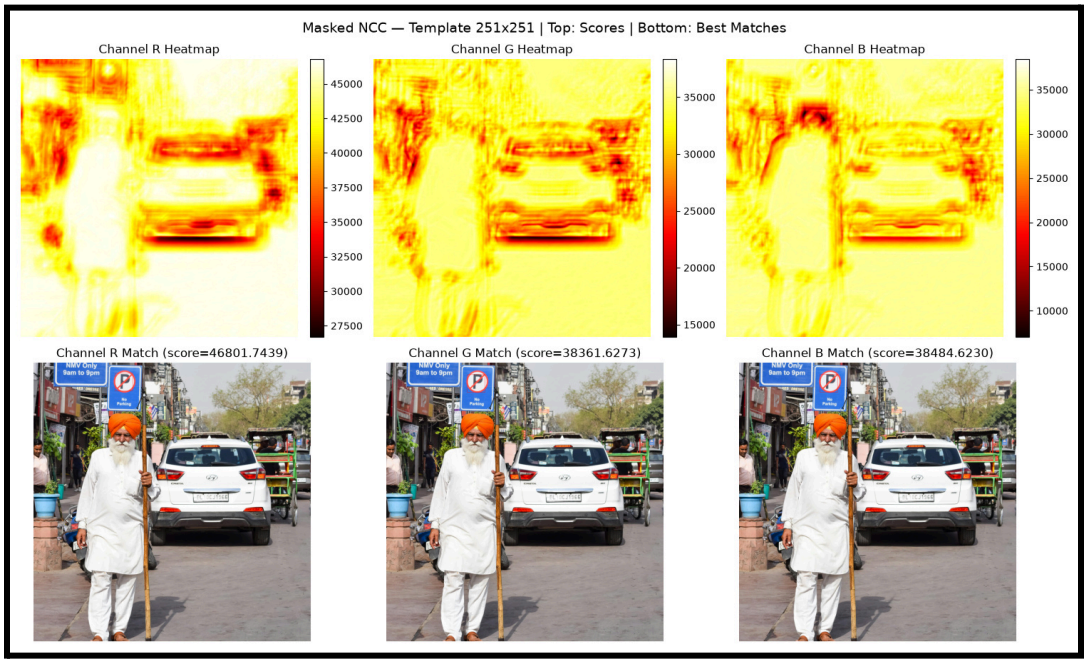
1. At 41x41 template size, The R channel locks onto the car's trunk, while the G and B channels lock onto the man's lower pant leg. The main usage of a smaller template is that, it can perfectly fit inside smaller & highly localized bright regions but is too small to accurately identify the match in the target image, especially in this case.

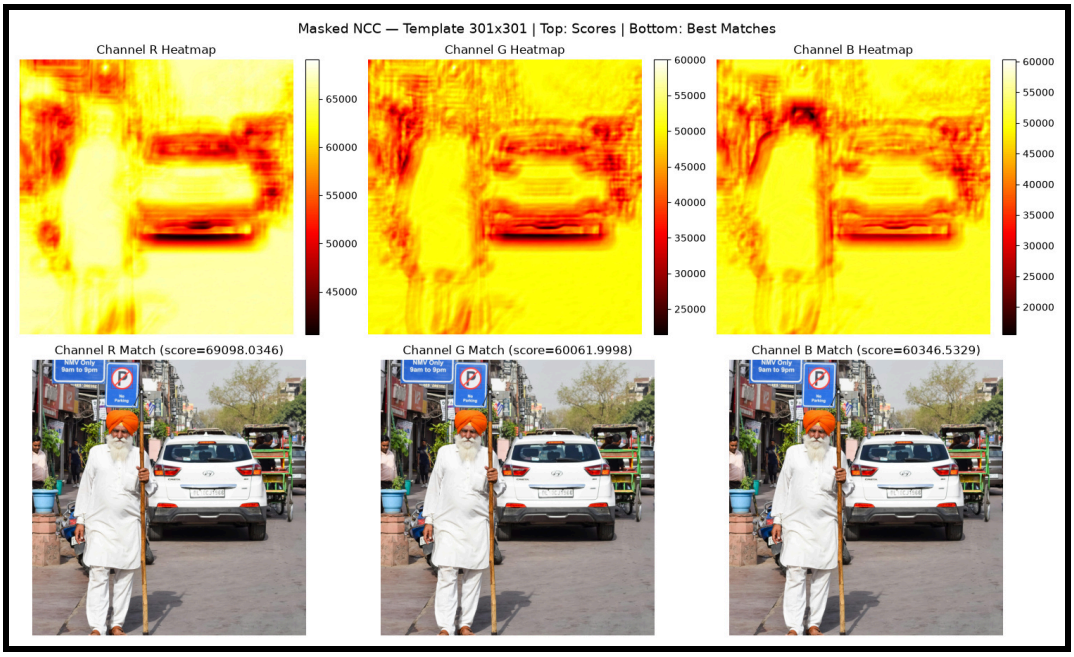
2. At 51x51 template size, the R channel targets the car's trunk, G channel targets the shadow but the B channel correctly identifies the no parking sign correctly. This signifies that increasing the template size helps the algorithm.

3. At 61x61 template size, the R channel targets the old man shirt but now the template size is big enough to detect the no parking sign in both G & B channels. This experiment shows that accurate template size matters in order to perform template matching on a given image.

Masked NCC


Masked NCC — Template 201x201 | Top: Scores | Bottom: Best Matches
Channel R Heatmap
Channel G Heatmap
Channel B Heatmap
Channel R Match (score=30025.1071)
Channel G Match (score=23769.9348)
Channel B Match (score=23904.8866)


Masked NCC — Template 251x251 | Top: Scores | Bottom: Best Matches
Channel R Heatmap
Channel G Heatmap
Channel B Heatmap
Channel R Match (score=46801.7439)
Channel G Match (score=38361.6273)
Channel B Match (score=38484.6230)



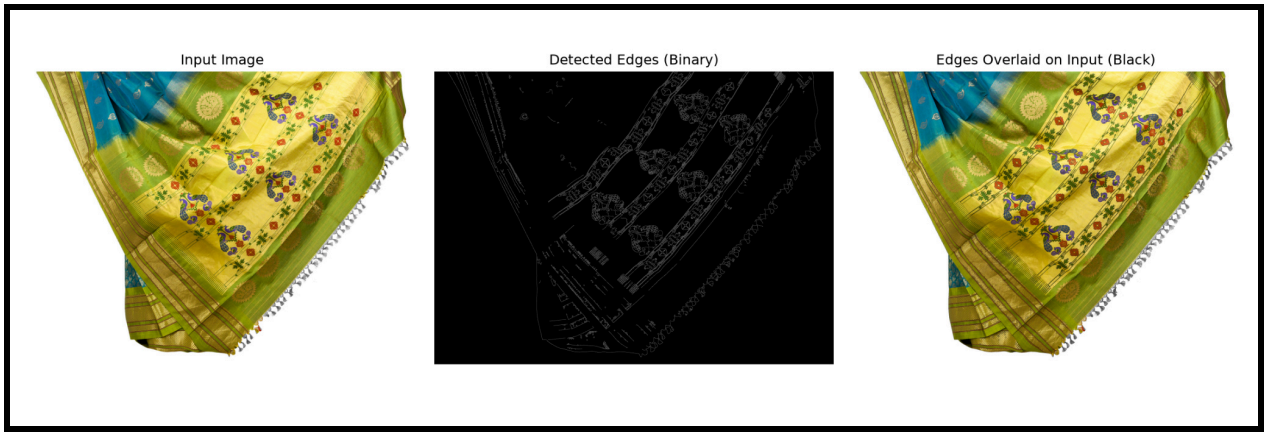
Explanation: When we apply masked template matching to the image:

1. At 201x201 patch size, the algorithm fails to detect even after using the masked NCC which might signify that the template might be too small for the algorithm to perform the operation.

2. However, when the template size is 251x251 & 301x301, the algorithm successfully identifies the no parking sign in the target size in all size images. This shows that masked NCC with accurate template size helps the algorithm to correctly identify the match.

3. Canny Edge Detection





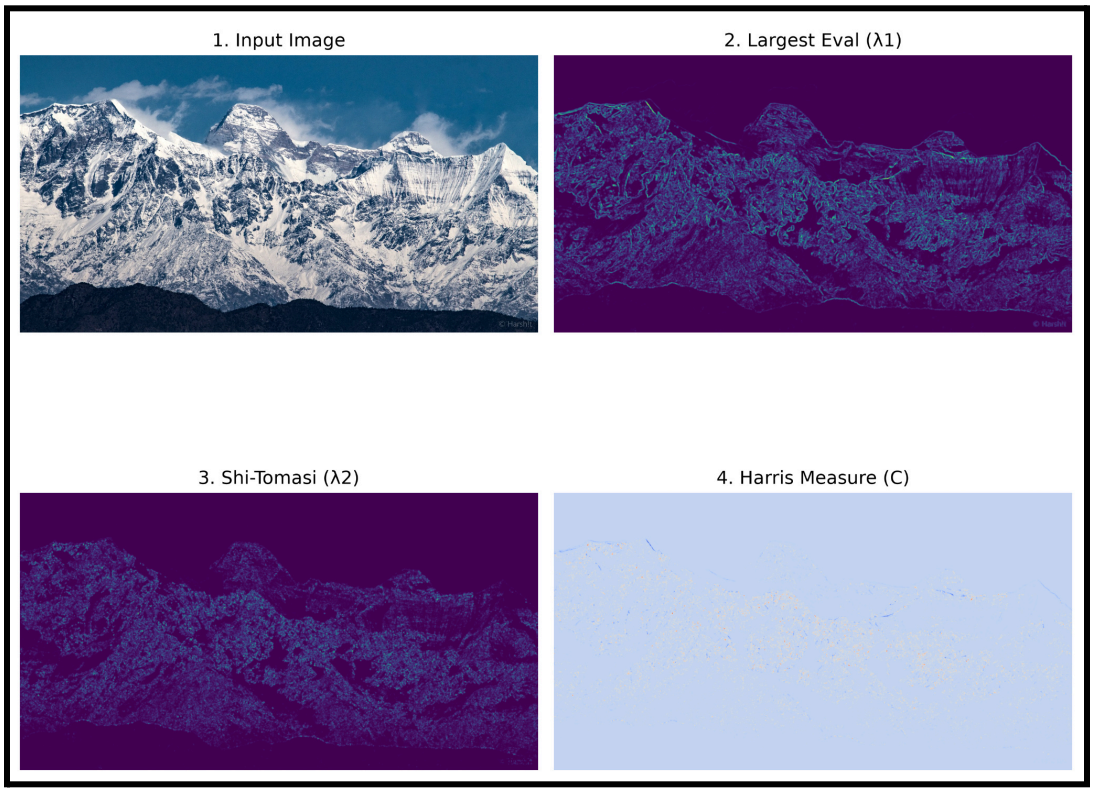


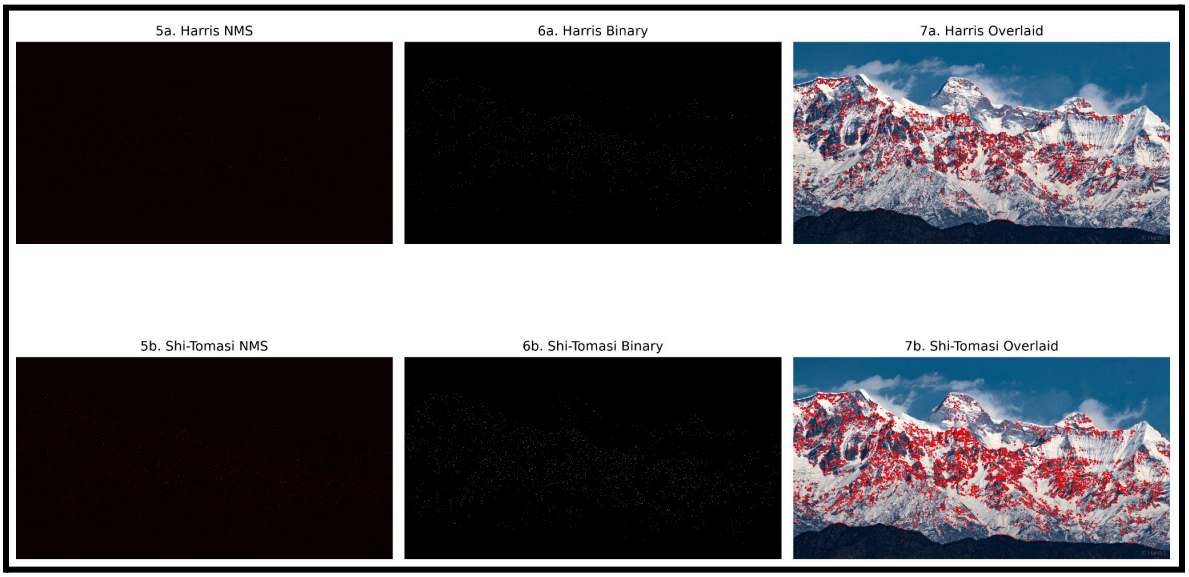
Explanation: When we apply Canny Edge Detection to the image:
1. butterfly.png (sigma = 1.0, low = 15, high = 30)
The low sigma value preserves the highly detailed structures of the delicate wing veins of the butterfly. The upper threshold (30%) is chosen so that it easily catches the bold black/orange borders while the lower threshold (15%) is chosen to ensure the intrication of inner wing patterns without the inclusion of the background noise.

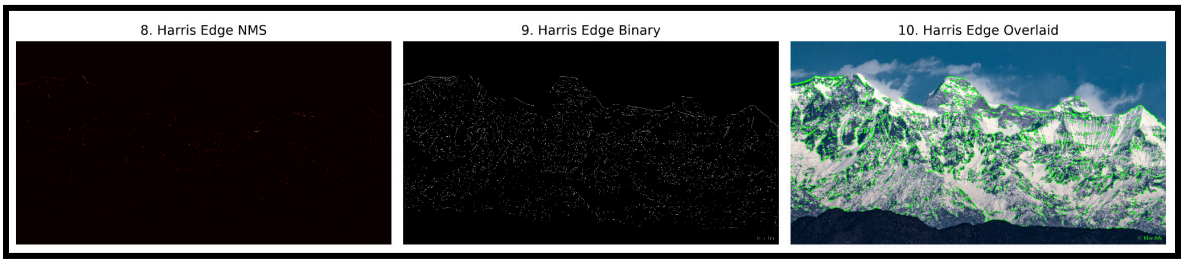
2. paithaniEdge.png (sigma = 1.6, low = 20, high = 45)
The fabric contains dense micro-textures which usually confuses the edge detectors with false positives and incorrectly considers such structures as noise. So, a gaussian filter with higher sigma value is used to mitigate the issue. The high threshold (45%) is chosen so that it ignores the fabric creases and grain present in the image.

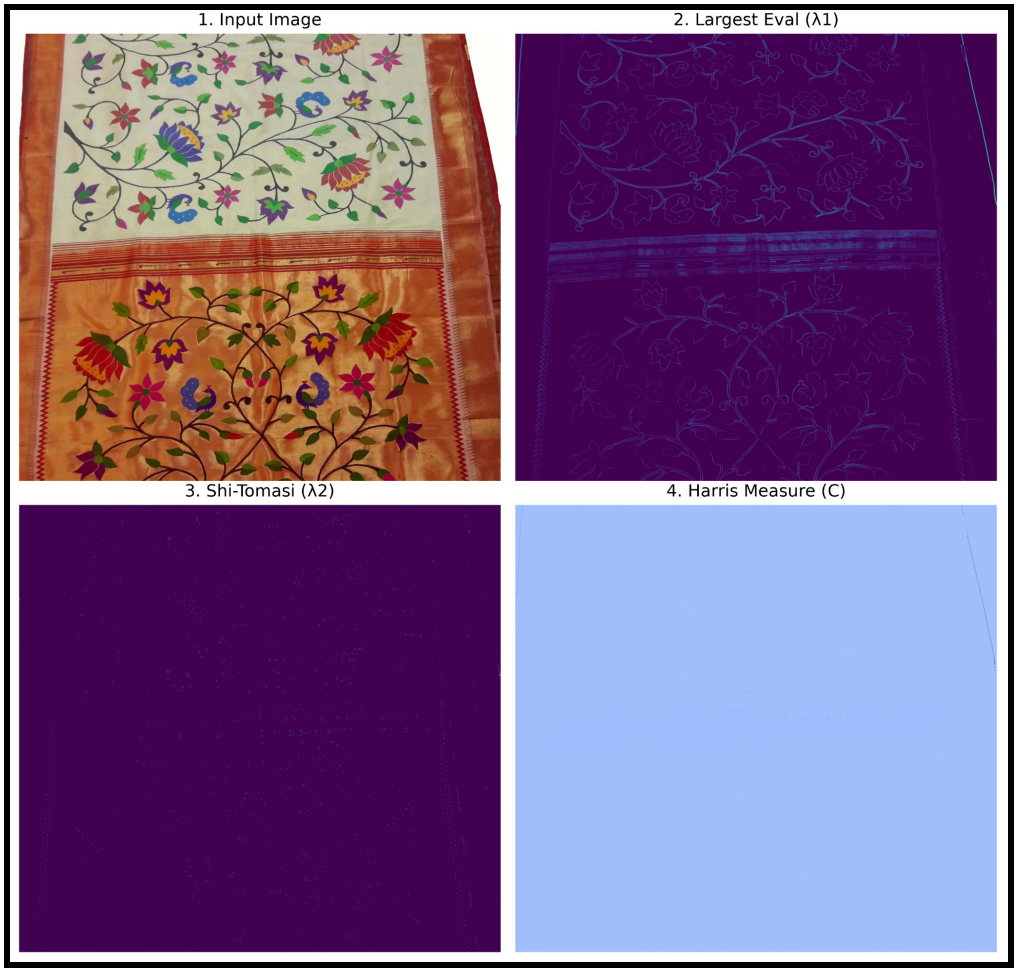
3. rangoli.png (sigma = 1.0, low = 10, high = 25)
The low sigma value here keeps the geometric chalk/powder lines sharp and distinct present in the rangoli image. The both thresholds are chosen in such a way that it allows the algorithm to detect the fainter gradients and successfully tracing the delicate yet also intricate loops extending into the shadows without breaking the edge geometry.

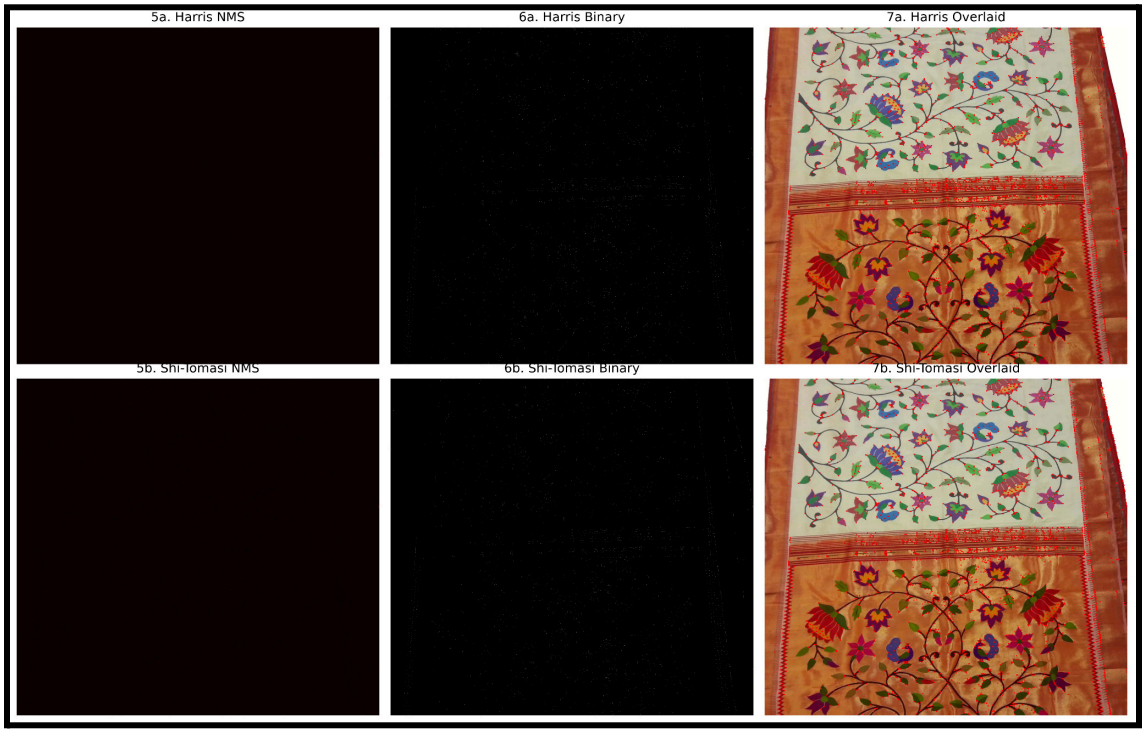
4. Edge and Corner Detection



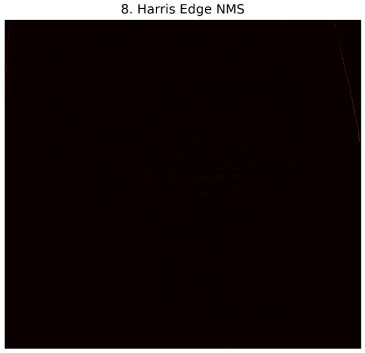




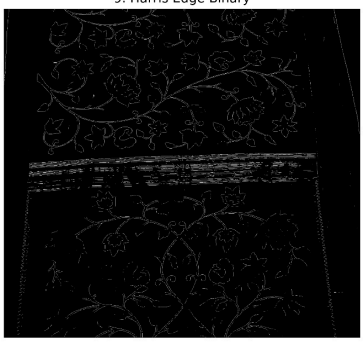

1. Input Image
2. Largest Eval (λ1)
3. Shi-Tomasi (λ2)
4. Harris Measure (C)


5a. Harris NMS
6a. Harris Binary
7a. Harris Overlaid
5b. Shi-Tomasi NMS
6b. Shi-Tomasi Binary
7b. Shi-Tomasi Overlaid

8. Harris Edge NMS



9. Harris Edge Binary



10. Harris Edge Overlaid



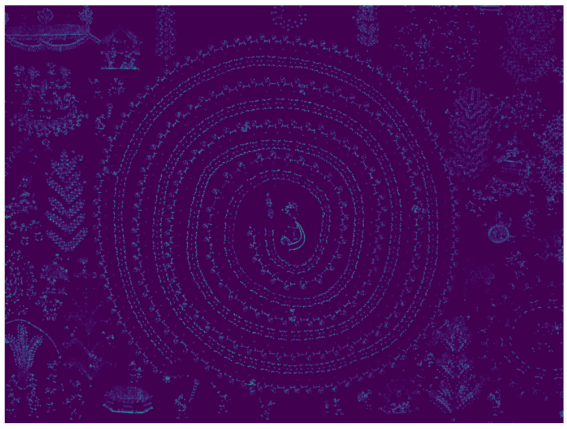
1. Input Image



2. Largest Eval (λ1)

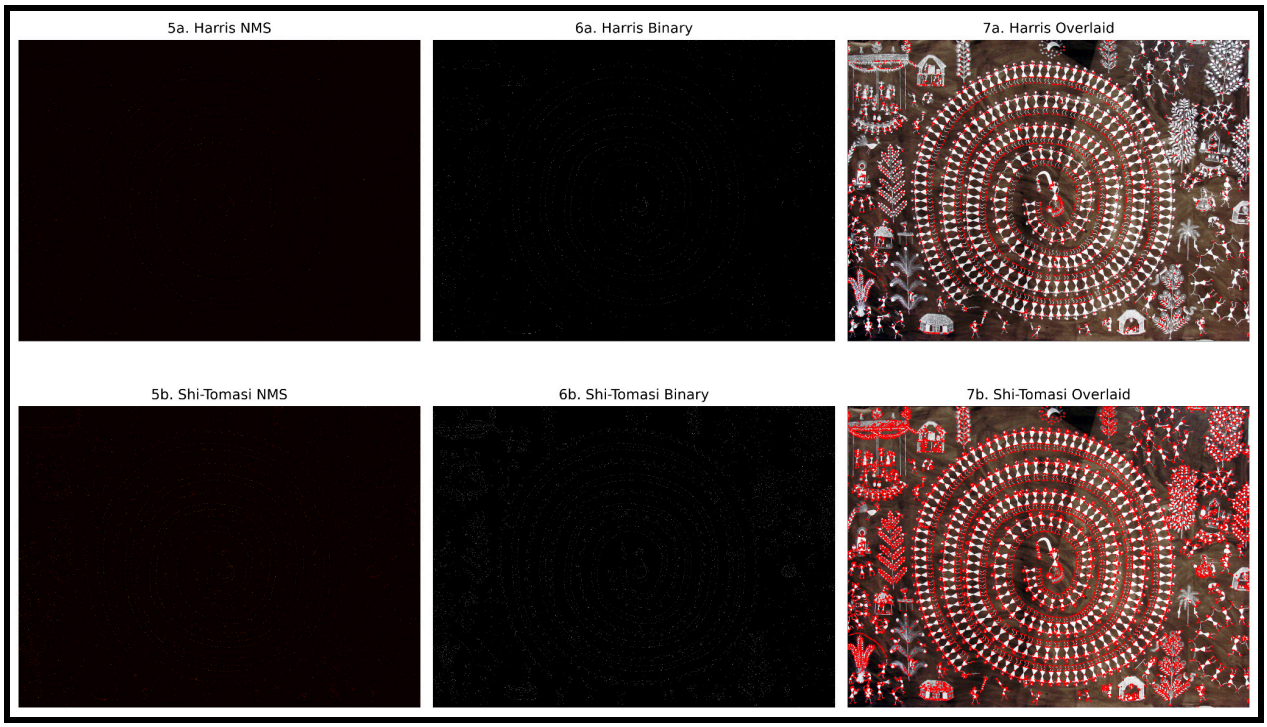


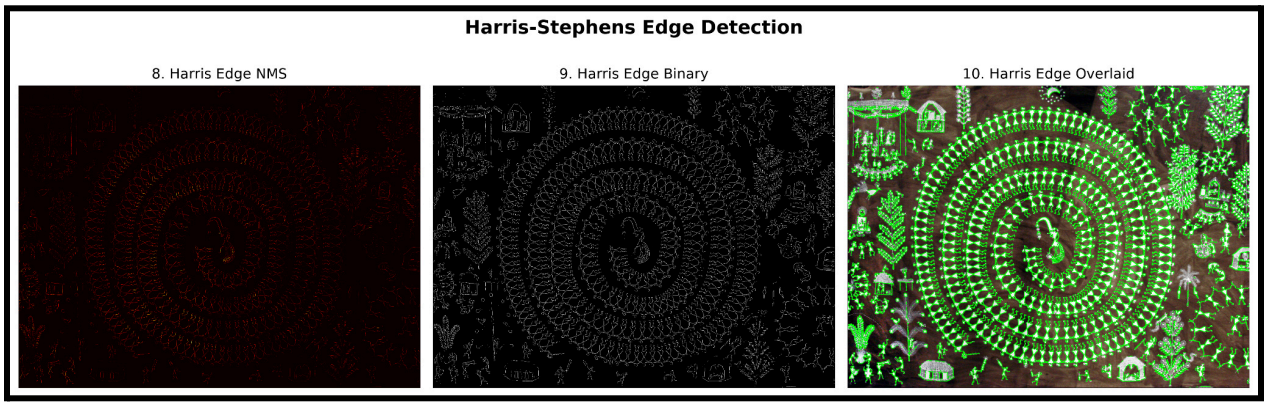
3. Shi-Tomasi (λ2)



4. Harris Measure (C)







Explanation: When we apply Corner & Edge Detection (Harris-Stephens & Shi Tomasi) to the image:

1. nandadevi.png

(k_corner = 0.08, c_thresh = 0.15, shi_c_thresh = 0.25, k_edge = 0.08, e_thresh = 0.015)

The value of k for both edges & corners is chosen so that it ensures the extraction of true structural ridgelines instead of just the boundaries of localized shadows. Pairing this with a lower threshold (for edges) & higher threshold (for corners in both Harris-Stephens & Shi-Tomasi) allows the topological lines to connect without breaking as well as it prevents the detector from flagging every single patch of textured snow as a corner.

2. paithaniCorner.png
(k_corner = 0.04, c_thresh = 0.001, shi_c_thresh = 0.03, k_edge = 0.06, e_thresh = 0.005)
The thresholds values chosen are microscopic because highly reflective metallic threads create massive, localized gradient spikes which skew the global maximum to an extreme value. Increasing the threshold values makes the algorithm not detect anything in the image. By dropping the fractional threshold down to 0.001, successful recovery of the legitimate geometric corners of the motifs takes place and the continuous edges get rescued from being overshadowed by the specular highlights.

3. warli.png
(k_corner = 0.20, c_thresh = 0.15, shi_c_thresh = 0.10, k_edge = 0.10, e_thresh = 0.02)
Keeping the high k_corner value removes any response from the long, continuous brush strokes which cleanly isolates the pure 2D intersections i.e. the heads of the figures and the sharp geometric joints. Keeping the same k value for edges would result in potentially suppressing the thinner brush strokes entirely; that's why k_edge value is chosen to be lower (0.10). Since the image is composed entirely of primitive 1D lines, dots & triangles, we observe cleaner results in this one.

5. Bokeh Effect





Explanation: When we apply Bokeh Effect to the images:
1. The lotus in lotus.png is perfectly isolated from its environment and the lily pads & pond in the background gets blurred. As the filter diameter is increased to D = 100, the water and lily pads smoothen out and it blends completely into a green gradient which forces the sharp edges of the petals to pop out.

2. The highly textured marigold flower in marigold.png is kept in sharp focus against a high frequency background of stems and leaves. The high diameter of the filter aggressively melts those jagged background leaves into soft & abstract washes of color which help us focus on the flower mainly.

3. The top metallic bowl of floating candles in deep.png is chosen as the sharp foreground while the lower arrangement of candles is blurred into the background. Applying the bokeh filters of both diameters transforms the sharp background flames into large, overlapping & circular highlights which gives a warmth to the image.